



HAZOP Leadership training for GC

Chapter 4 - HAZOP study methodology

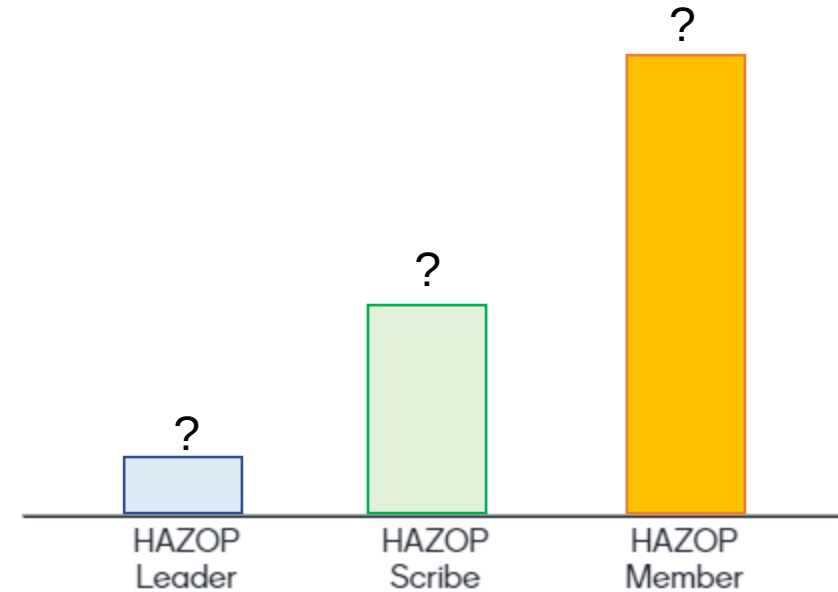
Technical Safety Service Division (Q-TS-TS), 1 - 3 November 2021

Activity-4 : Your Role in HAZOP team (Menti-4)

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What role were you in HAZOP study?

<https://www.menti.com/1g4hx55ryn>



❑ Internal Leader

- Minimum 8 years' experience in Refinery, Petrochemical, or Chemical process.
- Trained HAZOP Leadership course
- Trained Awareness of Process Safety in Design
- Must be independent person



Qualified
leader list

❑ External Leader

- External leader shall be qualified and registered by Q-TS-TS

Note

HAZOP coordinator shall submit the curriculum vitae (CV) to HAZOP Procedure custodian for approval.
Qualification of external leader are provided in HAZOP Study guideline

Outline of HAZOP study method

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- ☐ Overall Process Description (Process engineer or licensor)
- ☐ Selection of Nodes
- ☐ Detail description of design intent of node / system
- ☐ Identify Deviations (Apply Guideword – Parameter approach)
- ☐ Identify Causes (Within the node)
- ☐ Identify Consequences (Hazards & Operating Difficulties)
- ☐ Safeguards / IPLs
- ☐ Risk evaluation
- ☐ Recommendations / Actions

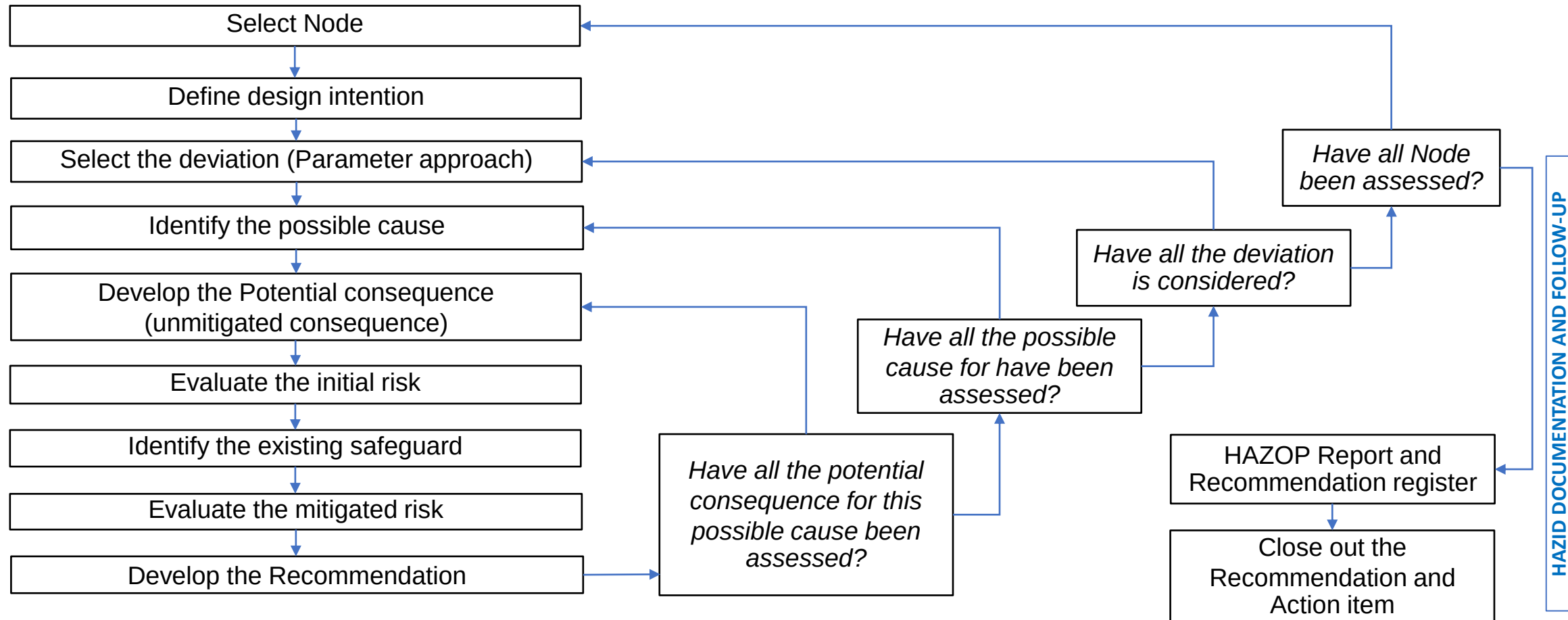
Outline of HAZOP study method

HAZOP Leadership Training Course

HAZOP PREPARATION

1. Define scope & objective study
2. Planning and Scheduling
3. Team selection
4. Agreement for Risk Assessment Matrix (RAM)
5. Process safety information and other support document preparing
6. Node Preparation

HAZOP METHODOLOGY



Chapter 4-1

Node preparation

Node selection and Design intent

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Outline of HAZOP study method

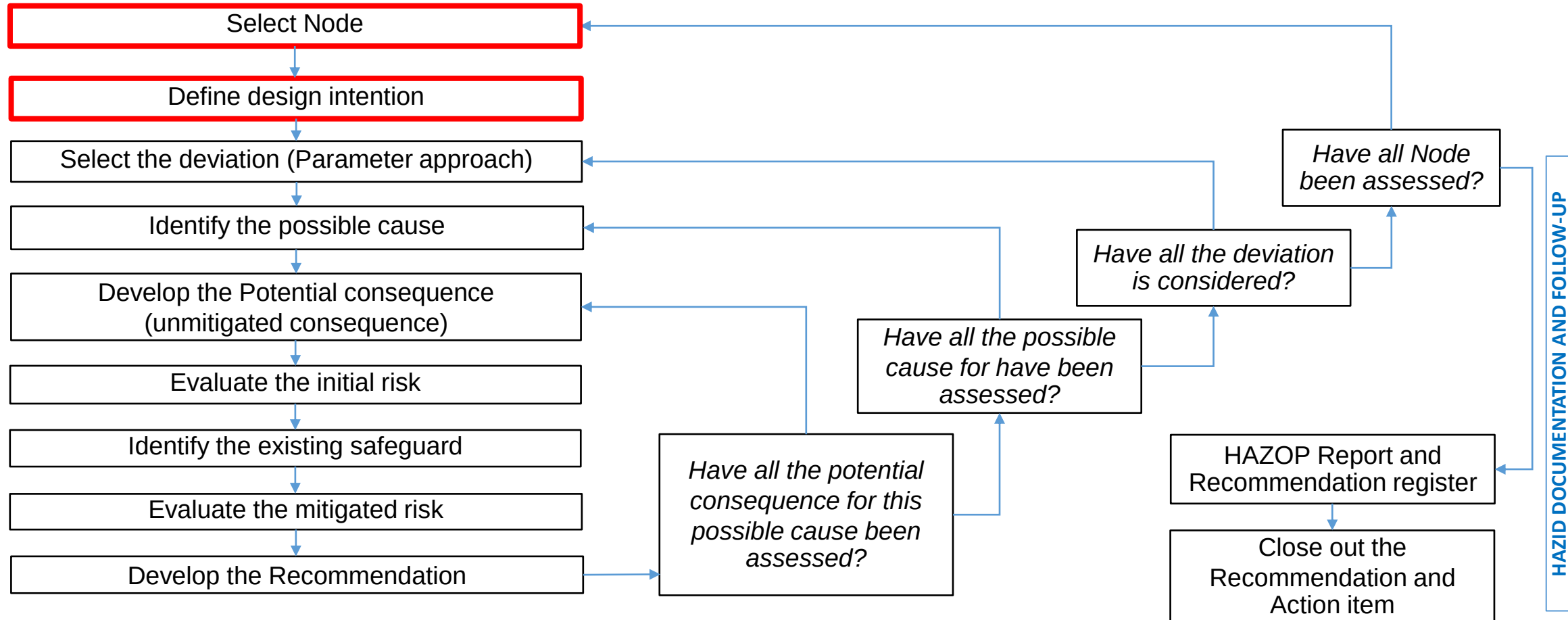
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HAZOP METHODOLOGY



Node is the breaking of the overall complex process design into a number of simpler sections which are then individually reviewed.

- ☐ Usually defined by the study leader prior to/or during the meetings in consultation with process engineer
- ☐ P&IDs are divided into nodes or sections of the process to be studied
- ☐ Logical process flow
- ☐ Nodes are points where the process parameters have an identified design intent
- ☐ Within the nodes are the components that cause changes in the parameters between nodes
- ☐ Some changes in nodes are to be expected as the study progresses
- ☐ Good practice guides often state that nodes should not cross from one P&ID to the next

- ❑ **Grouping of equipment with a similar purpose** (don't complicate design intents or mix parameter changes)

Strict: Whenever a parameter changes

Practical: End of transfer lines, at vessel interface

- Normally a single line or a vessel would constitute a node
- Linkage between equipment
- A short line bringing a feed stream into the process may be review together with the upstream interface
- Pressure control and relief lines can be included with a vessel
- Where lines pass through, for example, heat exchangers or condensers, these can be included in the same node
- A complete vapor circuit associated with a compressor, even though the system covers several flow sheets

- ❑ **As a study point**

- In each vessel or In transfer lines

- ❑ **Some items or lines have different functions at different stages of the process –**
e.g. regeneration of absorption beds

Separate node or re-studied by repeating the same node with different operation mode for system that have more than one operation mode (e.g. Regeneration, De-coke, Significant change in production type or spec),

- **Different service fluid or process condition**

Cause of the different inherent hazard and hazardous event of event in each operation mode

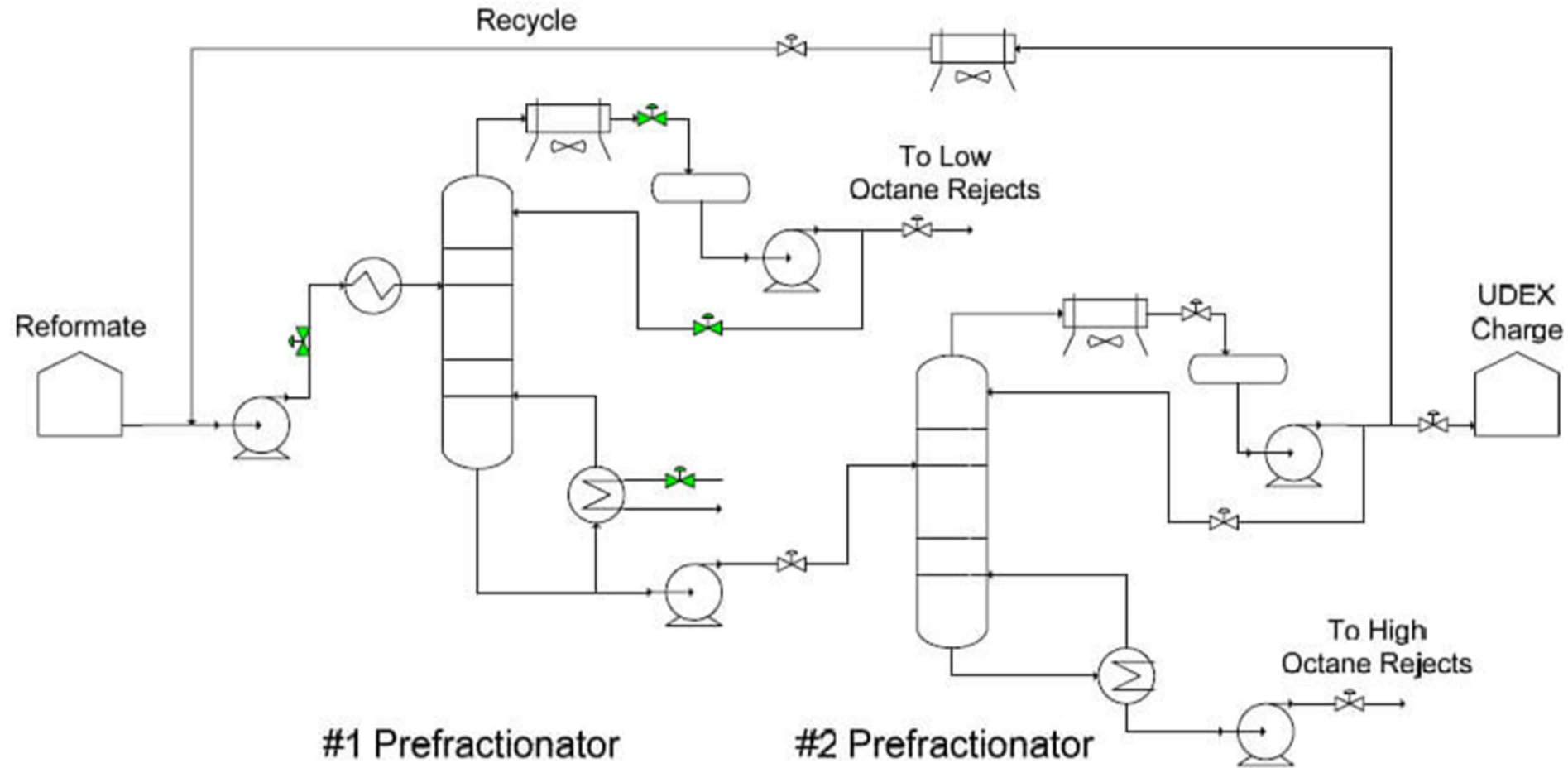
- **Different flow direction / line up / Equipment path**

Cause of different boundary of review and cause-consequence development in each operation mode

- ❑ If in any doubt, discuss the break down into nodes with the process engineer
- ❑ Share your proposed strategy with the team to make sure they are comfortable with it

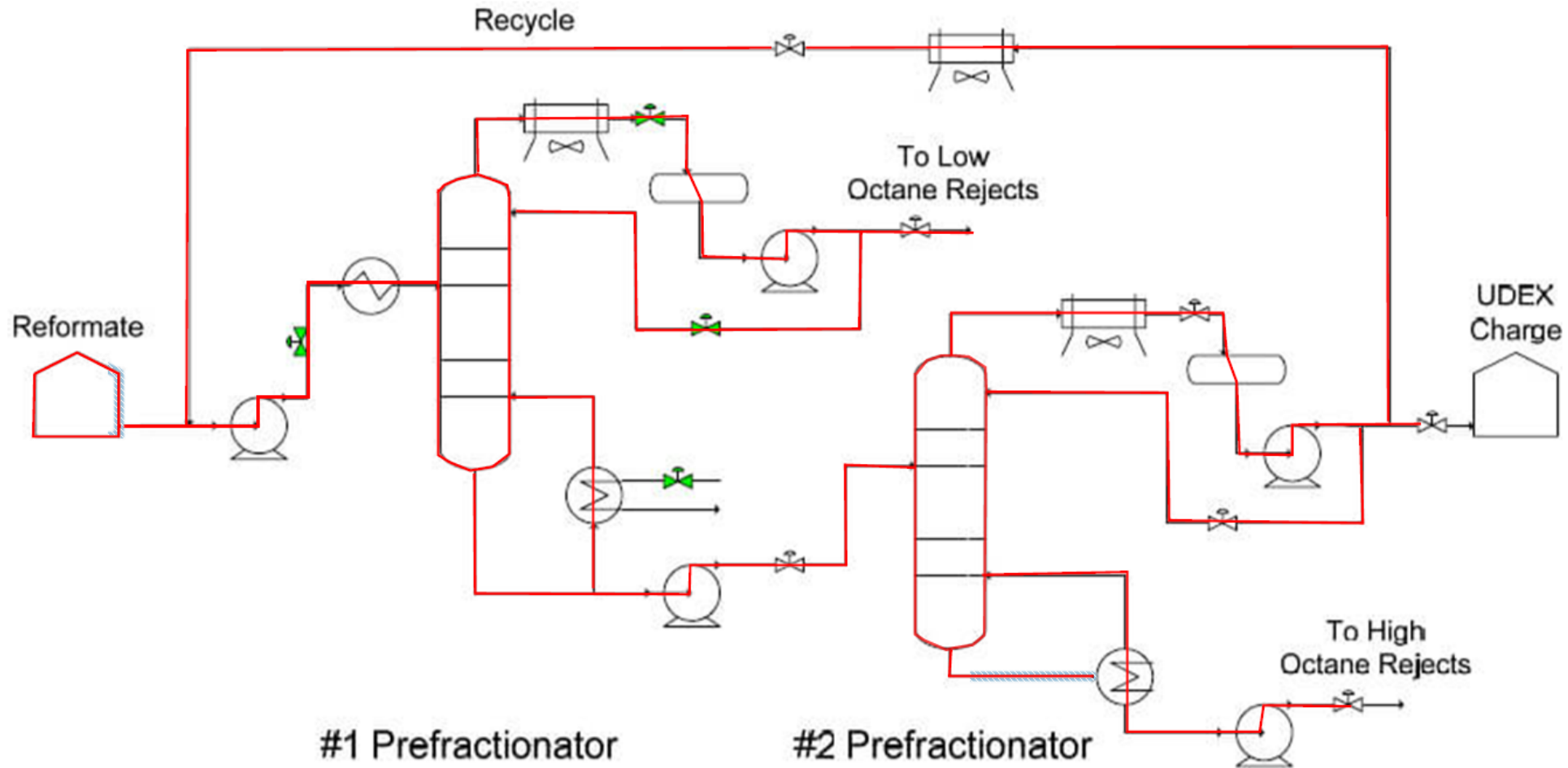
Node Preparation cont

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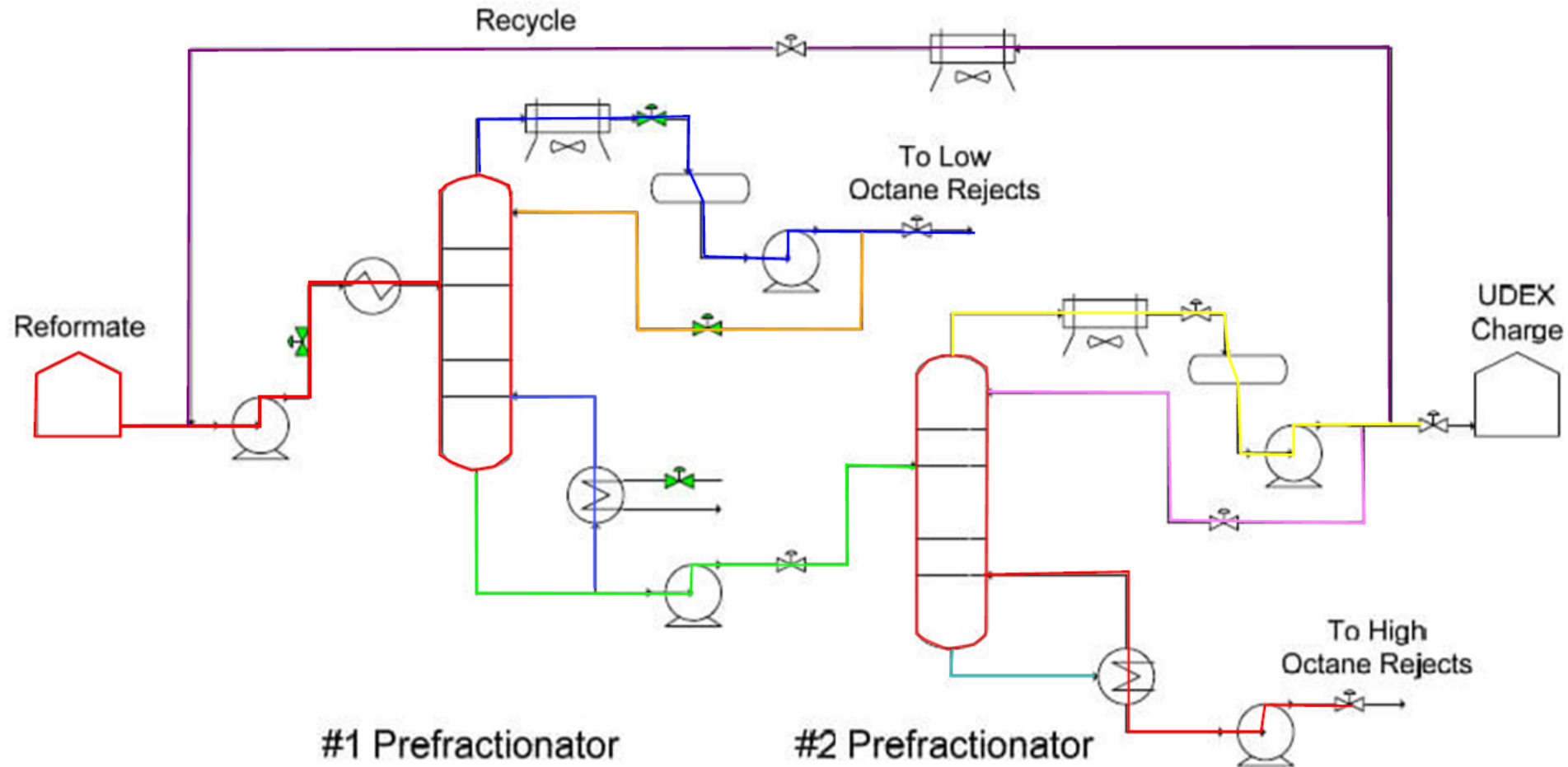
Node Preparation cont

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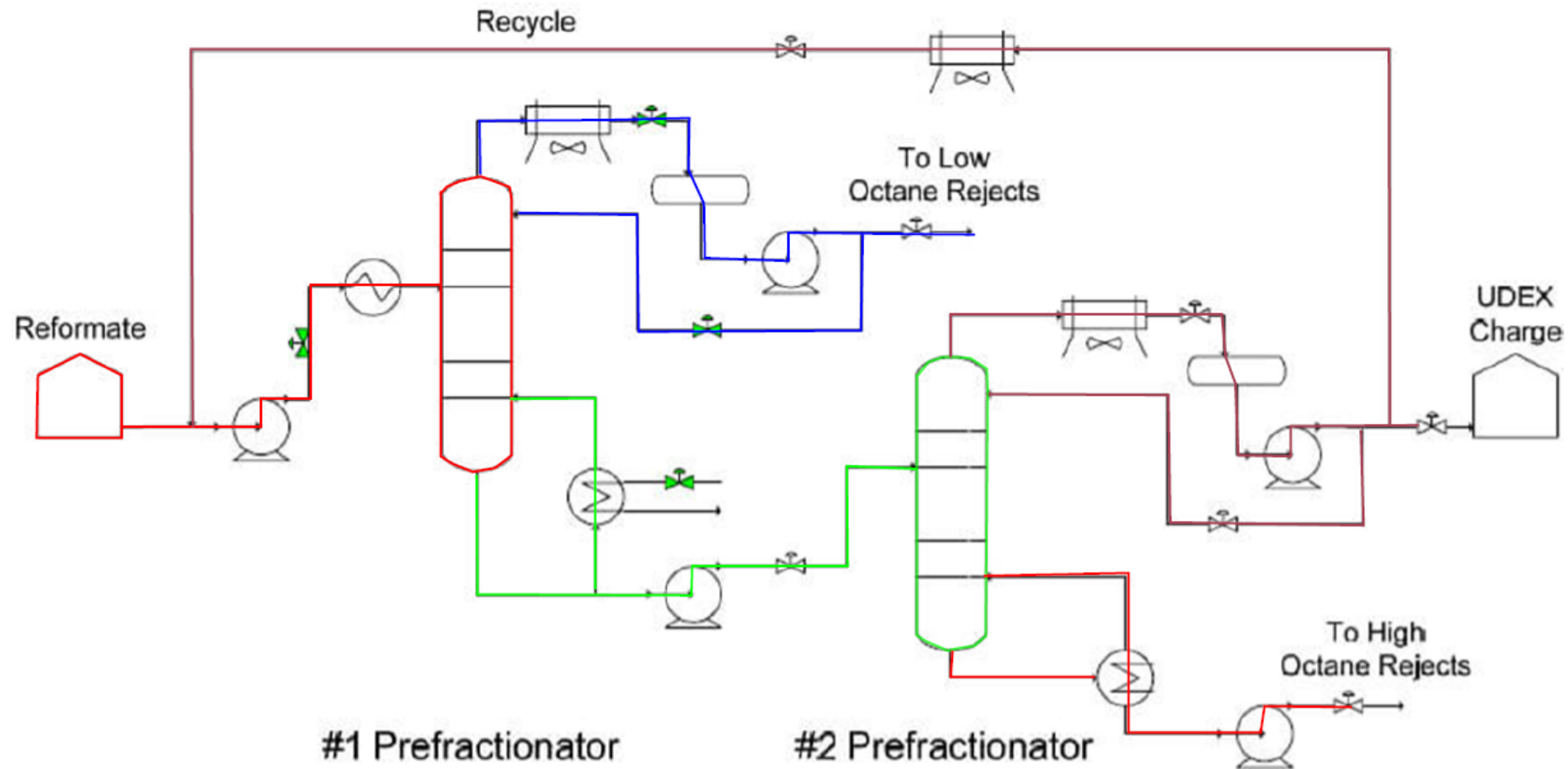
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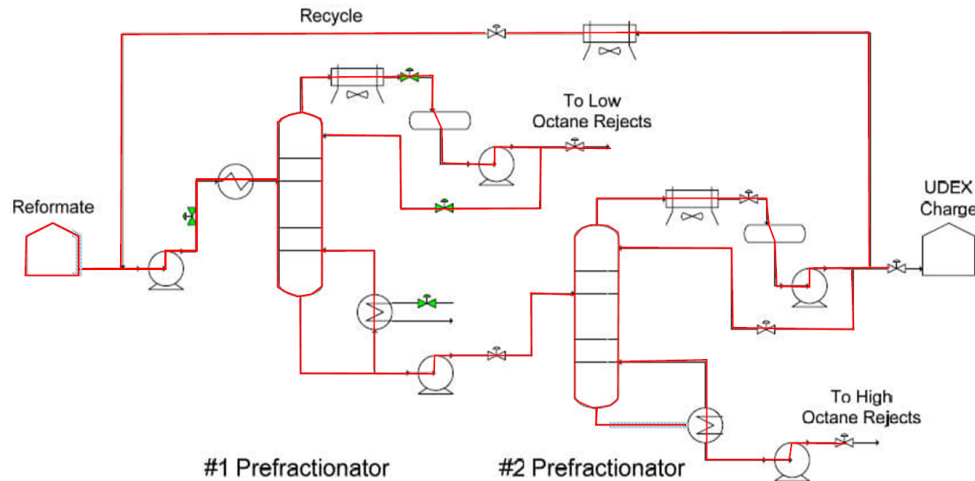
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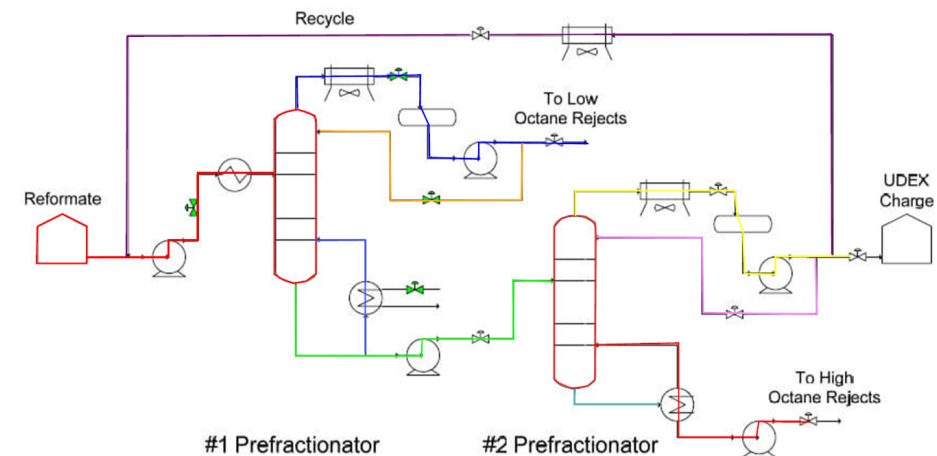
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Big node

- Save the time of study
- Hard to handle, lead to team confusing and potential miss identification for some of hazard



Small node

- Easy understand and identify possible cause
- Extremely time consuming
- Extensive repetition of recorded data
- Lead to extreme fatigue and loss of interest by HAZOP teams

Define Design/Operation Intent

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Set the “Operating” scene, and help people to understand the Node being studied

❑ Develop Design/Operation intent:

Design intention should indicate the following detail

- A description of normal design/Operation occurring at each study node
- Includes **functions** of node, stream compositions, and numerical value ranges of all parameters of importance
- Should say where stream come and go to
- Provide specific information e.g. exothermic reaction or endothermic (mildly, moderate, highly)
- Clear scope which not included in the node such as vendor packages, etc.

Example recording of design intent

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Node no. 1

Node description

Deethanizer overhead to acetylene converter, Acetylene Converter & C2 Dryer

Design intention

Gross-overhead deethanizer C-1301 is heated at E-1311A/B, E-1312 (LP steam, condensed to V-1312) and fed to Acetylene converter R-1301A/B for converting acetylene to ethylene (exothermic reaction). Effluent of Acetylene converter is cooled at intercooler E-1313, aftercooler E-1314. Then it is fed to ethylene dryer S-1304 and subsequently cooled at E-1311A/B, cold box A-1334X/1333X to Deethanizer Reflux drum V-1301.

Design condition

C-1301 design pressure = 18.5 kscg, design temp = -60/100 °C

V-1312 design pressure = 18.5 kscg, design temp = 260 °C

R-1301 design pressure = 18.5 kscg, design temp = -45/345 °C

S-1304 design pressure = 18.5 kscg (normal) 9.5 kscg (regen), design temp = -45/125 °C (normal) 280 °C (regen)

Operating condition

C-1301 Operating pressure = 14.9-15.2 kscg (13PC003 OVHD C-1301), Operating temp = -46 °C (OVHD C-1301)

V-1312 Operating level = 50% (13LC004)

R-1301 Operating pressure = 13.7 kscg (outlet R-1301), Operating temp = 60-116 °C (SOR to EOR at inlet temp),

Flow rate > 254,768 Nm³/hr

S-1304 Operating temp = 50-55 °C

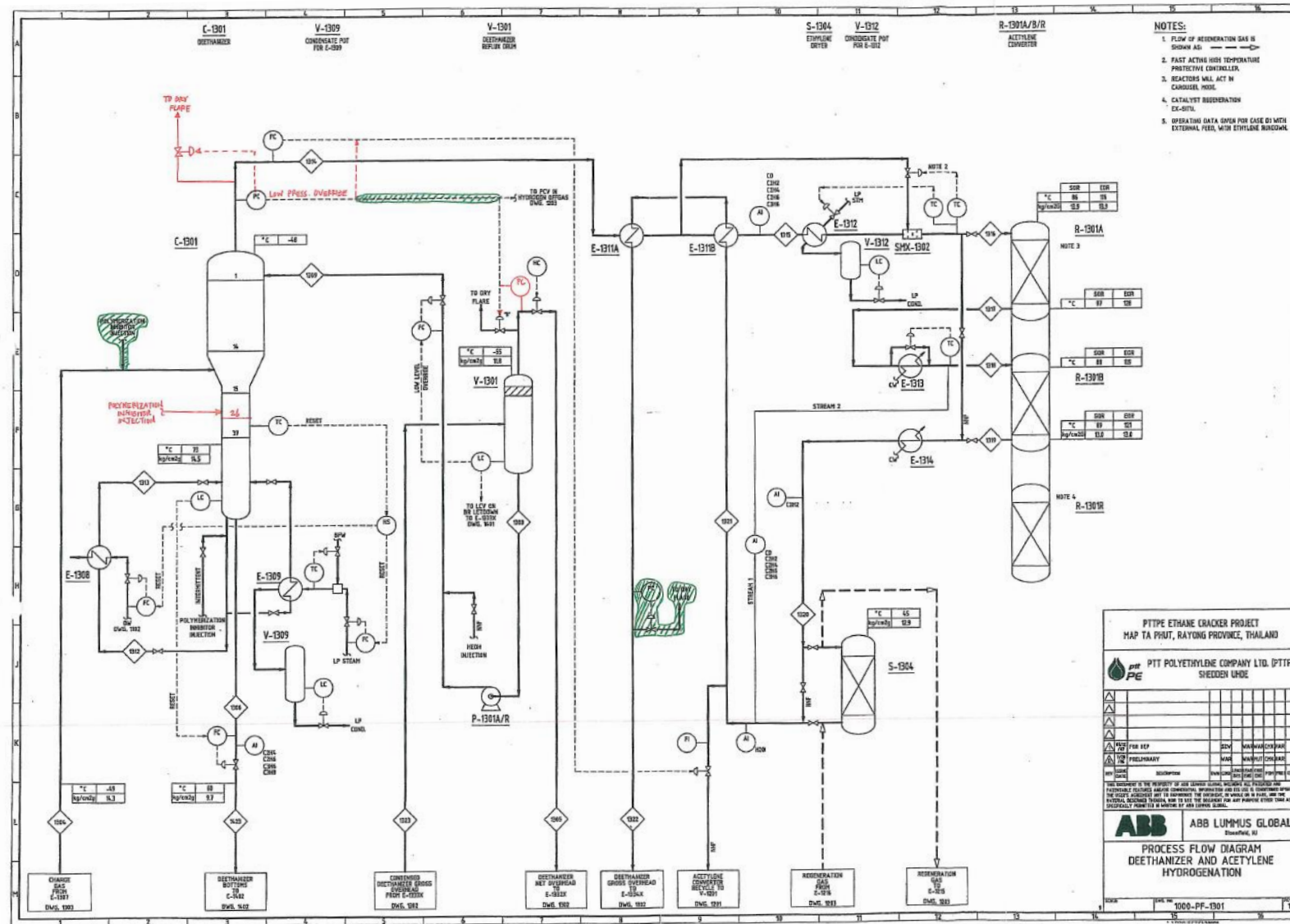
E-1313 Operating temp XXXX

A-133X Operating temp XXXX

Activity-5 : Node preparation (Brake group)

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Node preparation on PFD



- ❑ Group workshop 4 Group
- ❑ 15 min for node preparation
- ❑ 5 min for sharing (Each group)

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End of Day 1

Technical and Safety service (Q-TS-TS)
PTT Global Chemical PCL

Chapter 4-2

Guideword and Deviation

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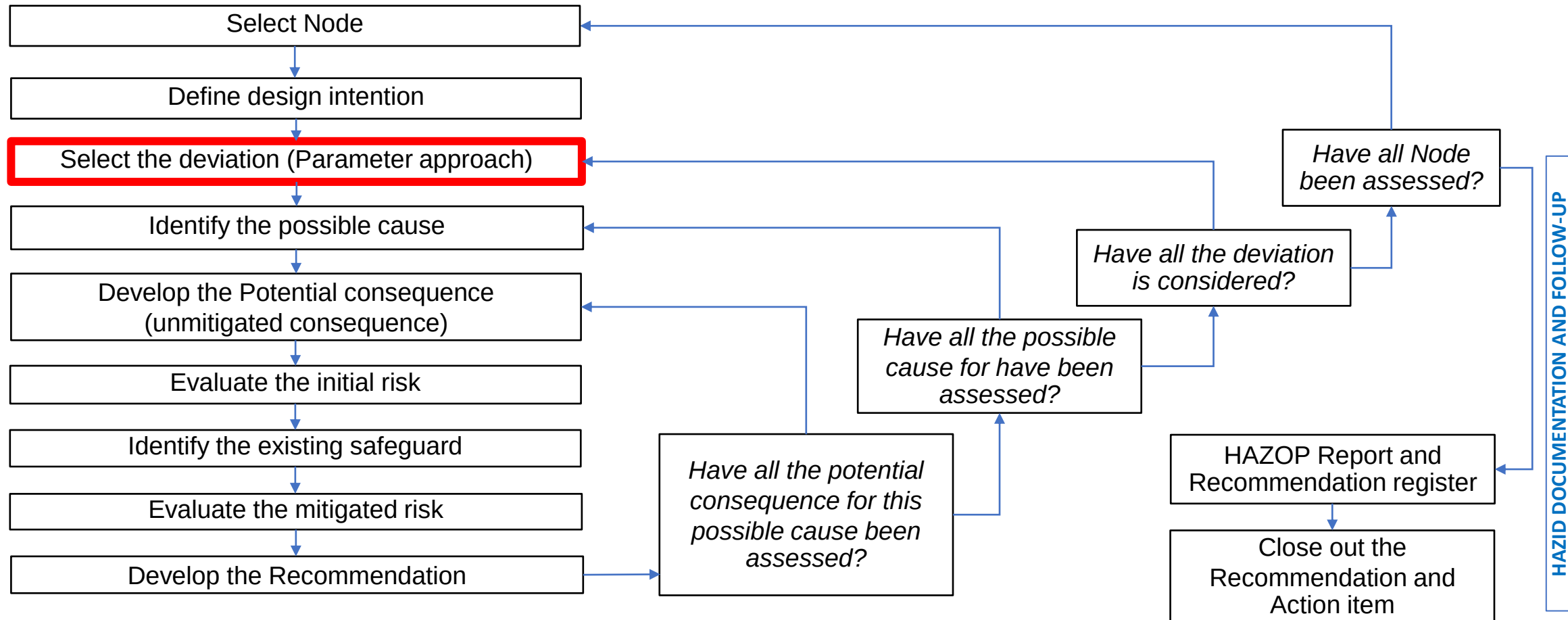
Outline of HAZOP study method

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HAZOP PREPARATION

1. Define scope & objective study
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HAZOP METHODOLOGY



Typical HAZOP Guideword

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Guide word	Meaning	Comments
No or NOT (None)	Complete negation of the intention	No part of the intention is performed, but nothing else happens
MORE	Quantitative increase	More or Less refers to - Quantities, e.g. amount , flow, Pressure, temperature, time - Properties, e.g. acid, alkaline, viscous - Functions, e.g. heating/cooling, Conversion
LESS	Quantitative decrease	
AS WELLAS	Qualitative increase	The intention is <u>achieved</u> , but <u>something else happens</u> <u>Further proceedings</u>, e.g. transfer & electrostatic load transfer to more than 1 location (only 1 specified) - More components in the system than should be, e.g. <u>additional phase</u> (vapor, solids in liquids), <u>contaminants</u>, <u>air</u>, <u>water</u>, <u>corrosion products</u>
PART OF	Qualitative decrease	Only some parts of the intention are performed - Multi-stage reaction takes an incomplete course - Partial or total missing of any material in a mixture - Particular system elements are missing or do not work

Typical HAZOP Guideword cont.

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Guide word	Meaning	Comments
INVERSION/REVERSAL	Logical Inversion of the intention	For proceedings: • Reverse flow (Opposite flow direction) • Contrarily running chemical reaction, e.g. decomposition instead of synthesis • Contrarily running physical process, e.g. condensation instead of evaporation For materials: • Inhibitor instead of catalyst
OTHER THAN	Complete Substitution, Deviation from Operating conditions. Other location	What else can happen? • Different material (raw material, intermediate, catalyst) • Different operation condition (Start-up, Shut-down) • Different chemical proceedings (undesired reaction) • Different physical proceeding (phase change) • Elsewhere

Common HAZOP Analysis Process Parameters

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Flow

Time

Frequency

Mixing

Pressure

Composition

Viscosity

Addition

Temperature

pH

Voltage

Separation

Level

Speed

Toxicity

Reaction

HAZOP Deviation

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Deviation

The combination of **Parameter** & **Guideword** which indicate departure from the design intent.

Note

All deviation should be considered and record.

		Guideword					
		No/none	More/High	Less/low	Reverse	Other than	Change
Parameter	Flow	No Flow	More Flow	Less Flow	Reverse flow	Misdirected Flow	
	Pressure		High pressure	Low pressure			
	Temperature		High temperature	Low temperature			
	Level	No level	High level	Low level			
	Reaction	No reaction	Runaway	Low reaction	Reverse reaction	Side/ Wrong reaction	
	Mixing	No mixing	Excessive mixing	Low mixing			
	Phase						Phase change
	Viscosity		High viscosity	Low viscosity			
	Composition						Composition change
	Erosion/ Corrosion		High corrosion/ erosion				
	Service failures	Loss of service	Excess of service				
	Sequence					Wrong step	
	Incidents					Previous/ Related incident	
	Human factor					Human factor	

Standard Set of Deviations

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ID No	Deviation	Column	Vessel	Line	Exchanger	Pump	Compressor
1	High Flow			X			
2	High Level	X	X				
3	High Interface		X				
4	High Pressure	X	X	X			X
5	High Temperature	X	X	X			X
6	High Concentration	X	X	X			
7	Low /No Flow			X			X
8	Low Level	X	X				
9	Low Interface		X				
10	Low Pressure	X	X	X			X
11	Low Temperature	X	X	X			X
12	Low Concentration	X	X	X			
13	Reverse / Misdirected Flow			X			X
14	Tube Leak				X		
15	Tube Rupture				X		
16	Leak	X	X	X	X	X	X
17	Rupture	X	X	X	X	X	X

CHECKLIST “Other”

- RELIEF SCENARIO
- INSTRUMENTATION
- SAMPLING
- CORROSION
- LEAKAGE
- SERVICE FAILURE

- MAINTENANCE
- START-UP
- SPARE EQUIPMENT
- SAFETY EQUIPMENT



Activity-6 : Identifying Deviations (Brake group)

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Refer to below Node 1 mark up
How many deviations do you applied?

Multiple answer

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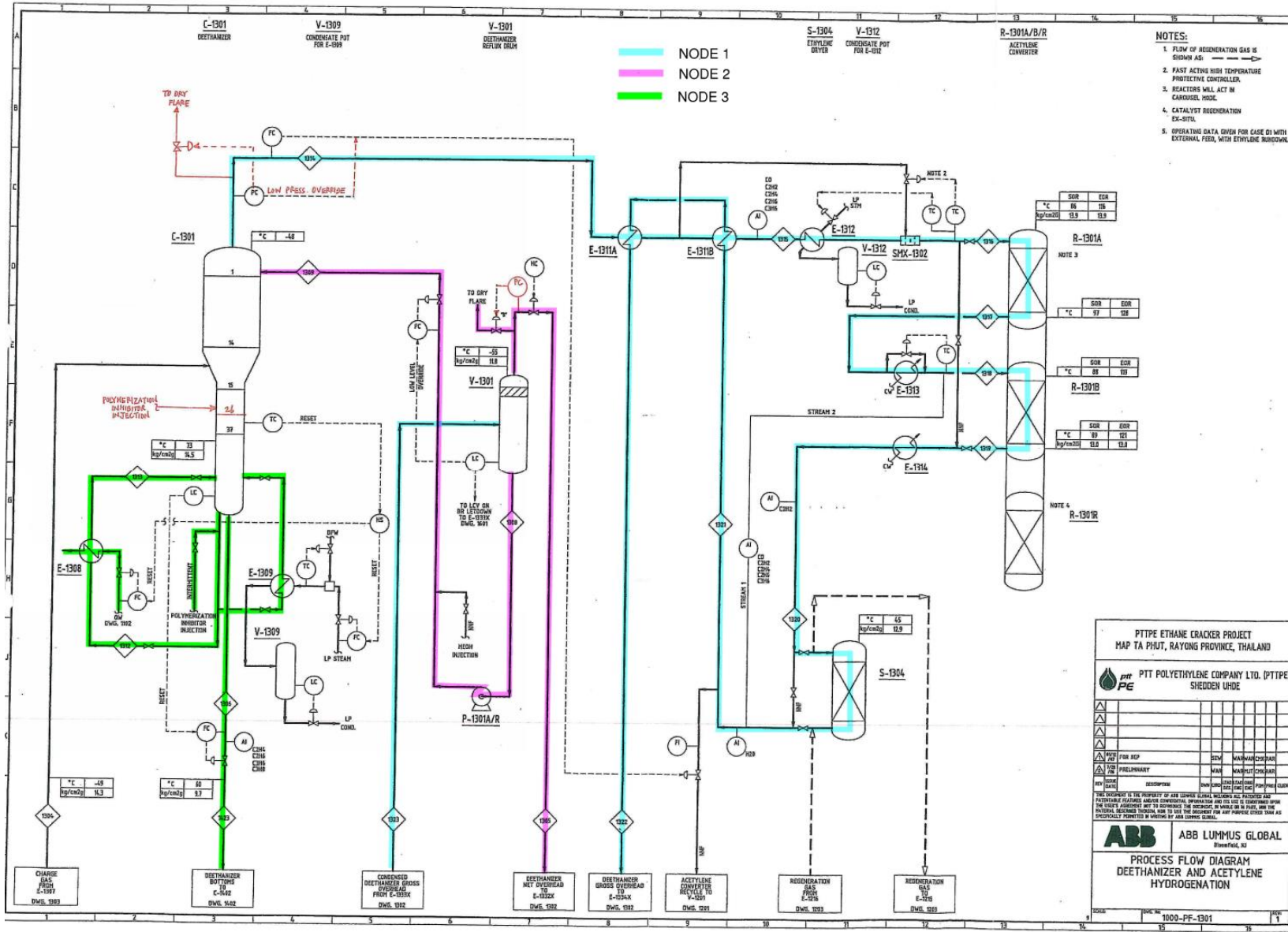
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- Sharing 5 min

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		Guideword					
		No/none	More/High	Less/low	Reverse	Other than	Change
Parameter	Flow	No Flow	More Flow	Less Flow	Reverse flow	Misdirected Flow	
	Pressure		High pressure	Low pressure			
	Temperature		High temperature	Low temperature			
	Level	No level	High level	Low level			
	Reaction	No reaction	Runaway	Low reaction	Reverse reaction	Side/ Wrong reaction	
	Mixing	No mixing	Excessive mixing	Low mixing			
	Phase						Phase change
	Viscosity		High viscosity	Low viscosity			
	Composition						Composition change
	Erosion/ Corrosion		High corrosion/ erosion				
	Service failures	Loss of service	Excess of service				
	Sequence					Wrong step	
	Incidents					Previous/ Related incident	
	Human factor					Human factor	

Activity-6 : Identifying Deviations (Menti-5)

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Node no. 1

Node description

Deethanizer overhead to acetylene converter, Acetylene Converter & C2 Dryer

Design intention

Gross-overhead deethanizer C-1301 is heated at E-1311A/B, E-1312 (LP steam, condensed to V-1312) and fed to Acetylene converter R-1301A/B for converting acetylene to ethylene (exothermic reaction). Effluent of Acetylene converter is cooled at intercooler E-1313, aftercooler E-1314. Then it is fed to ethylene dryer S-1304 and subsequently cooled at E-1311A/B, cold box A-1334X/1333X to Deethanizer Reflux drum V-1301.

Chapter 4-3

Developing and Recording of Possible Cause

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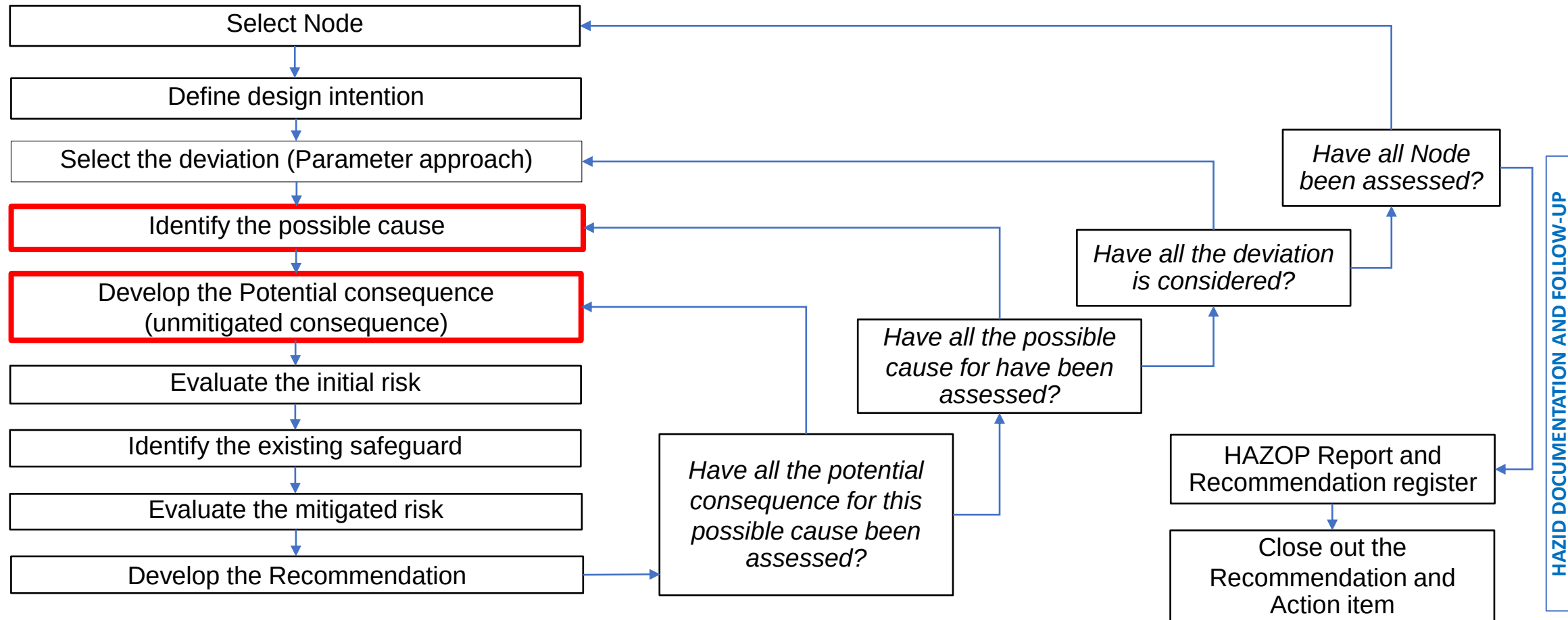
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HAZOP METHODOLOGY



HAZID DOCUMENTATION AND FOLLOW-UP

Identify the possible cause

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Possible cause is the reasons why deviations might occur

- ❑ Possible cause **shall be within the node being studied** while the potential consequence may occur throughout the process outside node
- ❑ **Exception:** Boundary nodes may consider upstream causes for completeness, but do not detail specifics of causes
- ❑ HAZOP Leader shall manage the brainstorming to ensure that team **not try to find the root cause or any solution** in HAZOP study

Recording of possible cause

- Clearly stated & Sufficient information for hazardous event to describe the ways in which each hazardous event can occur but Do not a restatement of deviations
- Using equipment tag, refer P&ID grid, refer prior equipment, refer destination equipment
- More than one possible cause may be identified in the one deviation which shall be recorded separately
- Record “No credible cause was/were identified”, “Not credible” or “Not applicable”

Example Possible Cause recording

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DO

- Malfunction of 11-TIC-012 leading to 11-TV-012 fully open
- No butene-1 supplied from T-6000
- Fitter at the inlet of A-4451 totally blocked
- Pump P-1101A/B stopped
- Drain valve at V-1301 left open



YES

DON'T

- Control Fault
- No feed flow
- Fitter blocked
- Pump faulty
- Low level in vessel



NO

Identify the possible cause

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Typical Causes considered under HAZOP



- BPCS instrument loop failure;
- Regulator failure;
- Exchanger leak;
- Pump/compressor tripped;
- Utility failure;
- Specific human errors
- General – Loss of feed from upstream / obstruction downstream (to cover interfaces)

Causes NOT considered under HAZOP



- Design issues / concerns
- PSV or SIF failure
- Pipe flange / gasket leak
- Closure of manual valves labelled as LO/LC or CSO/CSC

Chapter 4-4

Developing and Recording of Potential Consequence

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Develop potential consequences

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Potential consequence is the hazardous outcome statement that may include :

- ☐ Potential consequence may be inside or outside the node being review
- ☐ Fully identify for both of immediate and final consequence without considering of the safeguards
- ☐ Think forward to downstream when identified the consequence outside node (Backward for some case which depend on applied practice)

“HAZOP Leader must decide whether to include the consequences in the node or to refer the analysis to later or more suitable node. Whichever decision is made, it is important that the consequences outside the study node are fully covered”

Recording of possible cause

- Separately record the multiple consequences from the same possible cause
- Think through and document the “step by step” chain of consequence and using the linking word to describe the initial event link to final unmitigated consequence
- “No consequence was identified”, “No adverse consequence”, “No significant consequence” may be use if no consequence

Example Potential Consequence recording

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DO

- Reaction runaway causing of high pressure & temperature inside reactor (R-110) resulting reactor fail/rupture, potential LOPC fire and explosion.
- Low level of BFW (cooling side) in E-440 leading to high H₂S content in off gas at overhead V-4422 due to low performance of absorption as a result of high temperature tail gas to C-442, resulting in incomplete combustion potential SO_x over legal limit
- Low reflux to C-1302 leading high temp and pressure at overhead of C-1302 potential loss of ethylene to overhead and overpressure at C-1302 resulting to LOPC, fire & Explosion, single fatality



YES

DON'T

- R-110 Runaway
- Low level in E-4401 causing high temperature tail gas resulting in incomplete combustion
- Loss reflux to C-1302 leading high temp and pressure at overhead of C-1302



NO

Chapter 4-5

Existing safeguard

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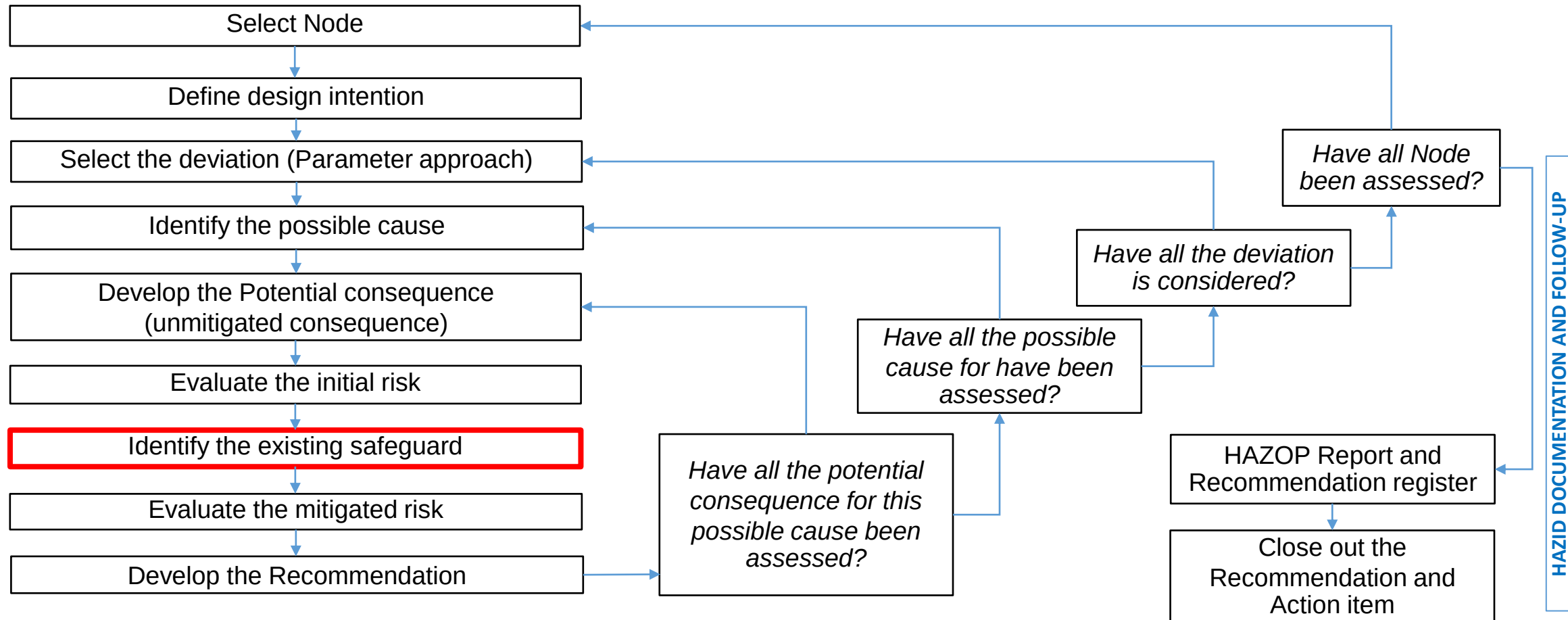
Outline of HAZOP study method

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HAZOP PREPARATION

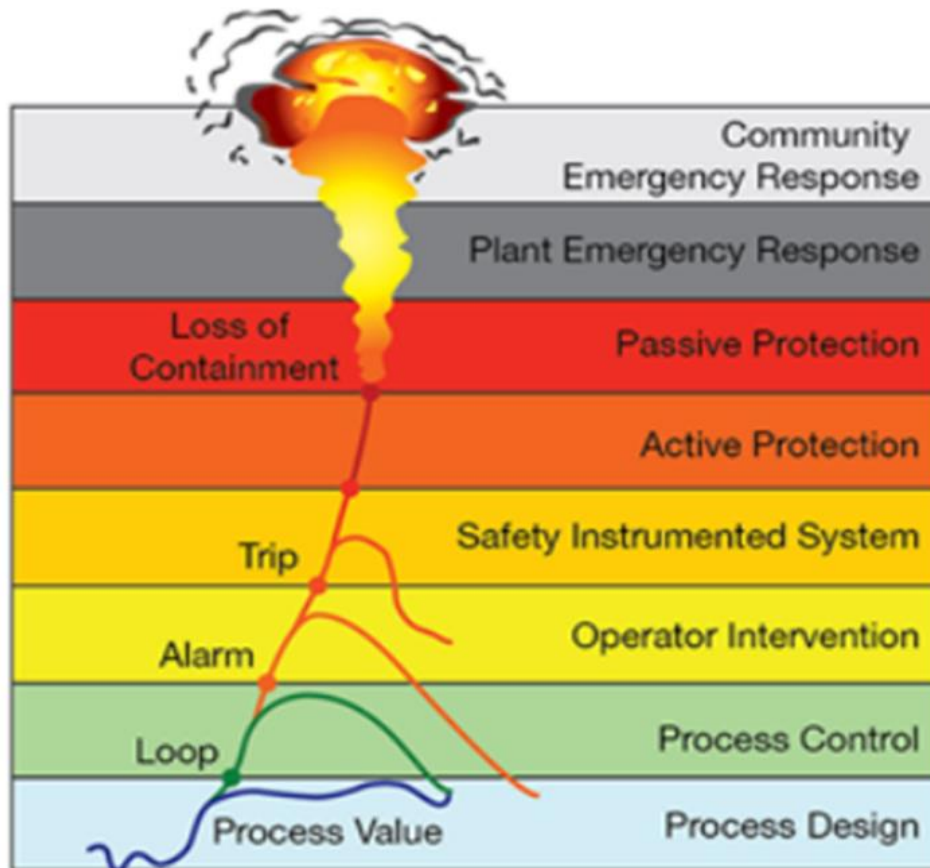
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HAZOP METHODOLOGY



Existing Safeguard

The engineered systems or administrative controls designed to prevent or mitigate the consequences that already present in facility or design information



Safeguards are not restricted inside of the node

Existing installations - shall presents in the local and document

Facilities under modification - shall presents in the design basis/philosophy or the technology of licenser, PFD, P&ID or any engineering document depend on the project state

Recording of potential consequence

- Records the prevention safeguard the first and, then mitigation safeguard. HAZOP leader should lead team by applying protection layer's concept
- General active fire protection and emergency response may be discussed in HAZOP study but are not listed as the safeguard
- Using equipment tag or location to identify the device.
- Describe the detail of safeguard to understand What protect and How to protect and protective condition if applicable.

Example Existing Safeguard recording

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DO

- Pressure control 13-PIC-100 (5 kg/cm2g) on separator V-1000
- High high pressure 14PAHH006A/B/C (10.7 KSCG, 2oo3 voting) to trigger Interlock Z-141 (SIL-1) to cut steam to reboiler E-1406A/R
- 11-PSV-056A/B/R (Set at 23 barg) on HP separator V-1000



DON'T

- High pressure alarm PAH
- PSV-100
- High level alarm
- High level alarm and trip



Typical Safeguard Preventive

- BPCS instrument loop e.g. PIC, FIC, TIC;
- Alarm with operator response e.g. PAH, PAL, LAH, LAL etc.;
- Safety instrument function (SIF),
- Mechanical switch:
- Pressure relief device e.g. PSV, Rupture disc, Emergency hatch;

Typical Safeguard Mitigative

- Dike, Impounding basin

Chapter 4-6

IPLs Concept

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Independent protection layer (IPL)

a device, system, or action that is **capable of preventing a scenario** from proceeding to its undesired consequence **independent of the initiating event or the action of any other layer of protection** associated with the scenario

All IPLs are safeguards, but not all safeguards are IPLs.

The safeguard that considered as the IPLs shall satisfy the following criteria to be considered valid

Effective

Fit to purpose
Capable to prevent the hazardous event

Independent

Not part of initiating event or another safeguard and Not
cause of new additional hazard

Auditability

Management systems that maintain barrier health

Effective

Fit for purpose and
Reliable

Can detect the hazard condition?

Can detect the condition in time to take corrective action?

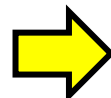
Does the IPLs have adequate ability for it to take the required action in the time available?

Example evaluation

Think about your previous HAZOP

Ex: Alarm with operator response layer

- How many alarms that you record as the safeguard?
- How many alarm that was considered for risk reduction
- Have a discuss about response time and action to prevent hazardous event



Example issue for discussion notice point

What parameter to detect the hazard?

- Is it appropriate parameter for scenario being reviewed?

Where is sensor located?

- Is it appropriate to detect that parameter?

What the setting value?

- Is it appropriate for operation to act with limited time?

What is the operator action?

- Is it can prevent the hazardous event?
- Is the complicate action or require?
- Is the action can do so without being put into a dangerous situation to accomplish the action?

Safeguarding with IPLs concept

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Independent

From initiating event
and other IPL

Ensure that the effects of the **initiating event**, or of **other IPLs**, do **not interact** with a specific **IPL** and thereby **degrade its ability** to perform its function.

See P&ID 1000-PI-1301

Example issue for evaluation

Ex: Alarm with operator response

Cause :

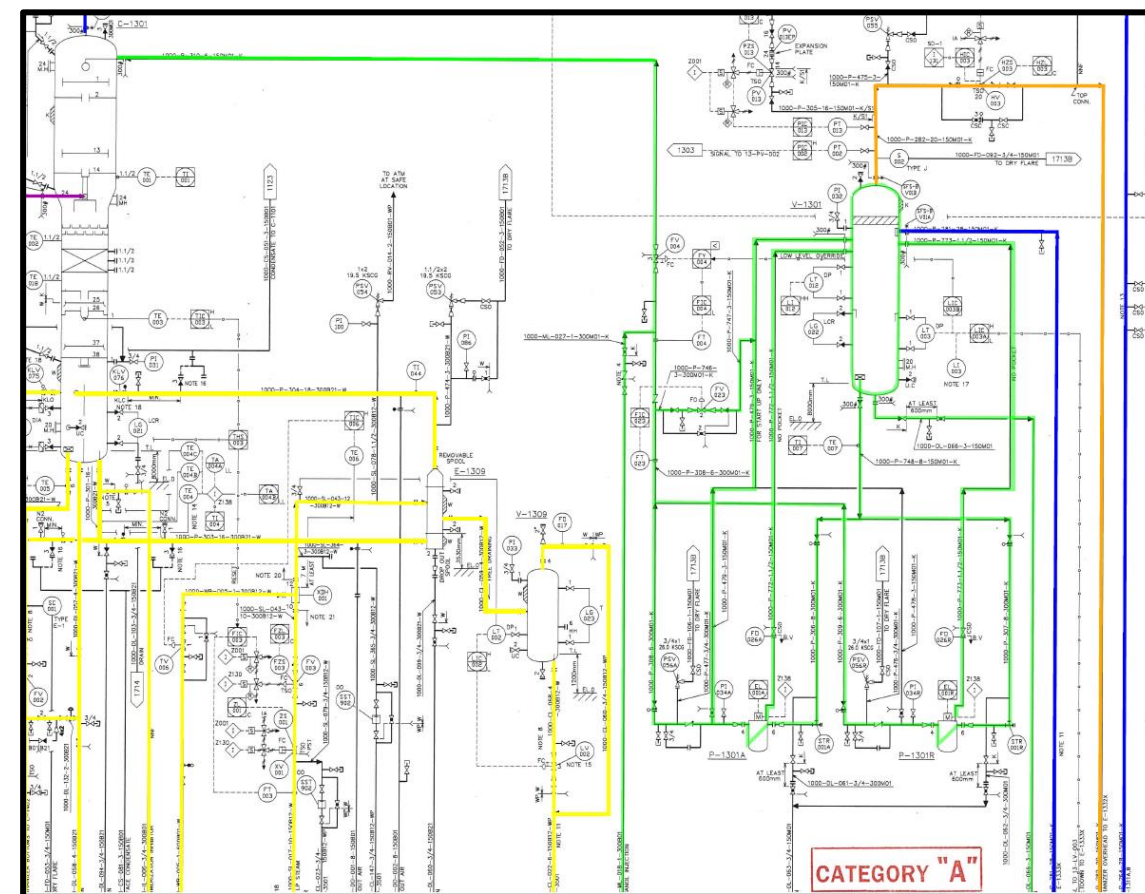
Malfunction of level control loop 13LIC003B leading to closing of 13FV004

Potential consequence :

High level liquid in V-1301 causing partial block inlet and low flow to R-1301 leading to hotspot and runaway causing fire and explosion

Safeguard : *(Independent????)*

1. Level control loop 13LIC003A to adjust amount of condense liquid to V-1301
2. High level alarm 13LAH003A with operator response
3. High-high level alarm 13LAH012 with operator response



Auditability

Management systems that maintain barrier health

To ensure that IPLs can be relied upon to operate as intended, under stated conditions, for a specified time period

Can be audited to ensure that the management systems to support the IPL are in place and effective.

“Please beware that HAZOP session is not audit, it not required to check more detail of evident, just confirm by related team”.

Rule/Assumption for HAZOP may be made to assume that company has the well management system and comply with standard.

Example issue for evaluation

Ex: SIL layer

There is evident for scenario for SIL assessment, Target and Verification

Confirm that there is management comply with standard e.g., Proof test within is proper period

Independent protection layer IPLs consideration

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Some safeguard that not consider as IPLs

CCPS Layer of Protection Analysis Simplified

Training and Certification or Procedure

These factors may be considered in assessing the PFD for operator action but are not of themselves IPLs.

Sign

Signs by themselves are not IPLs. Signs may be unclear, obscured, ignored, etc.

Normal Testing and Inspection and Maintenance

These activities are assumed to be in place for all hazard evaluations and form the basis for judgment to determine PFD. Lengthening the testing and inspection intervals may increase the PFD of an IPL

Fire Protection and emergency response

Active fire protection is often not considered as an IPL as it is post event for most scenarios and its availability and effectiveness may be affected by the fire/explosion which it is intended to contain.

Communications

It is a basic assumption that adequate communications exist in a facility. Poor communications affects the PFD of certain IPLs.

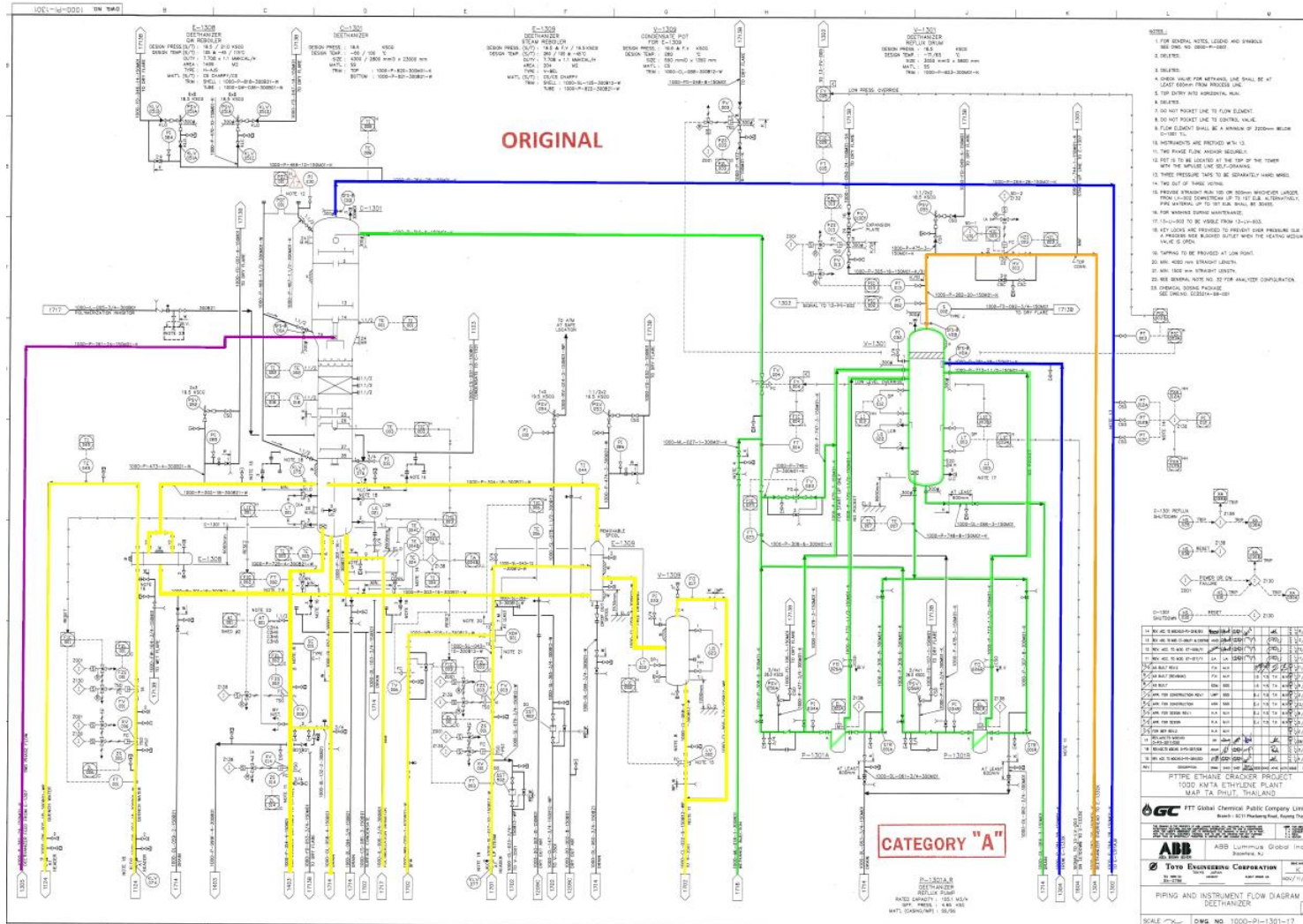
Requirement that Information is Available and Understood

This is a basic requirement

Activity-7: Identify the Safeguard and IPL (Brake group)

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Yellow node



Possible cause 1

Malfunction of 13TIC003 to fully open 13FV003

Potential consequence

More LP Steam to E-1309 leading to overboiled at C-1301, resulting in overpressure in C-1301, LOPC,

Identify Existing safeguard

Identify IPL

Possible cause 2

Malfunction of 13FIC003 to fully open 13FV003

Potential consequence

More LP Steam to E-1309 leading to overboiled at C-1301, resulting in overpressure in C-1301, LOPC,

Identify Existing safeguard

Identify IPL

Chapter 4-7

Risk Determination

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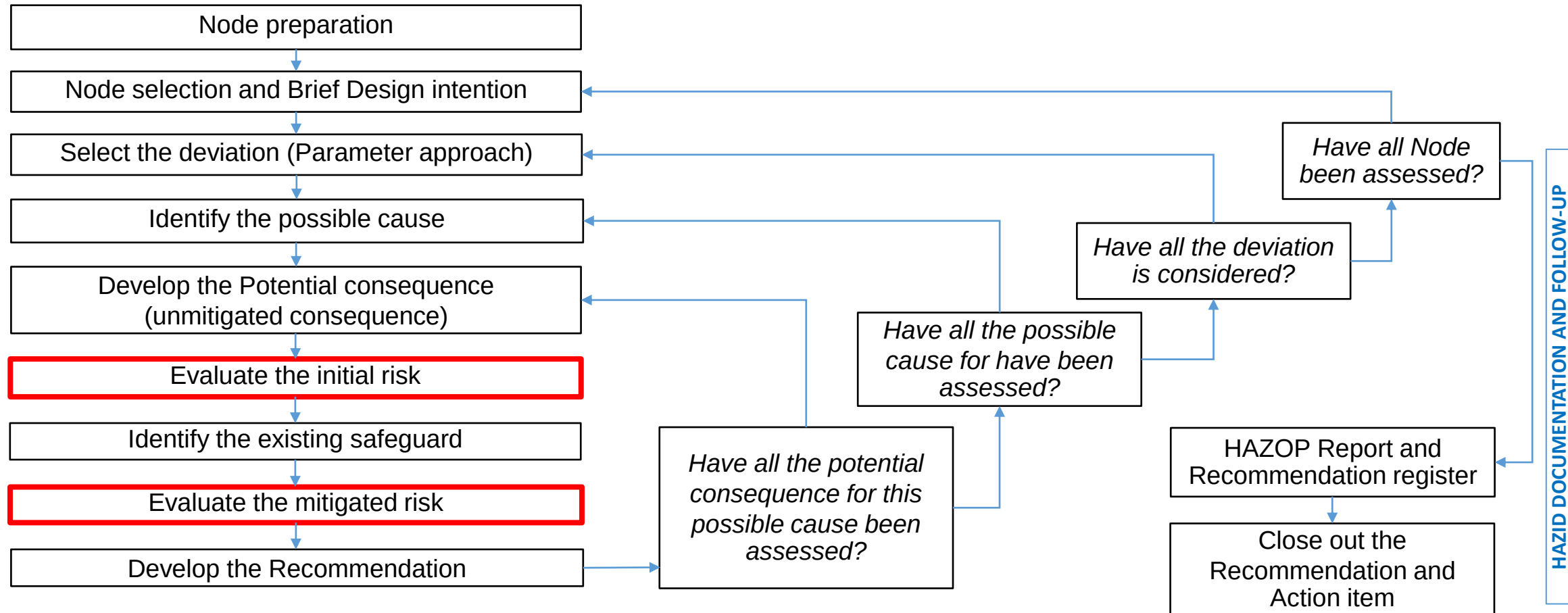
Outline of HAZOP study method

HAZOP Leadership Training Course

HAZOP PREPARATION

1. Define scope & objective study
3. Planning and Scheduling
4. Team selection
5. Agreement for Risk Assessment Matrix (RAM)
6. Process safety information and other support document preparing

HAZOP METHODOLOGY



HAZID DOCUMENTATION AND FOLLOW-UP

Example GC HAZOP Worksheet

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Parameter	Deviation	Possible Cause	Potential Consequence	Without Safeguard							Existing Safeguard	With Existing Safeguard							Recommendation
				L	Severity					RR	Detail	L	Severity					RR	Detail
					P	En	Ec	S	P				En	Ec	S				

Risk ranking Concept

Likelihood definition

Frequency of occurrence of unmitigated consequence.

Presenting of safeguard

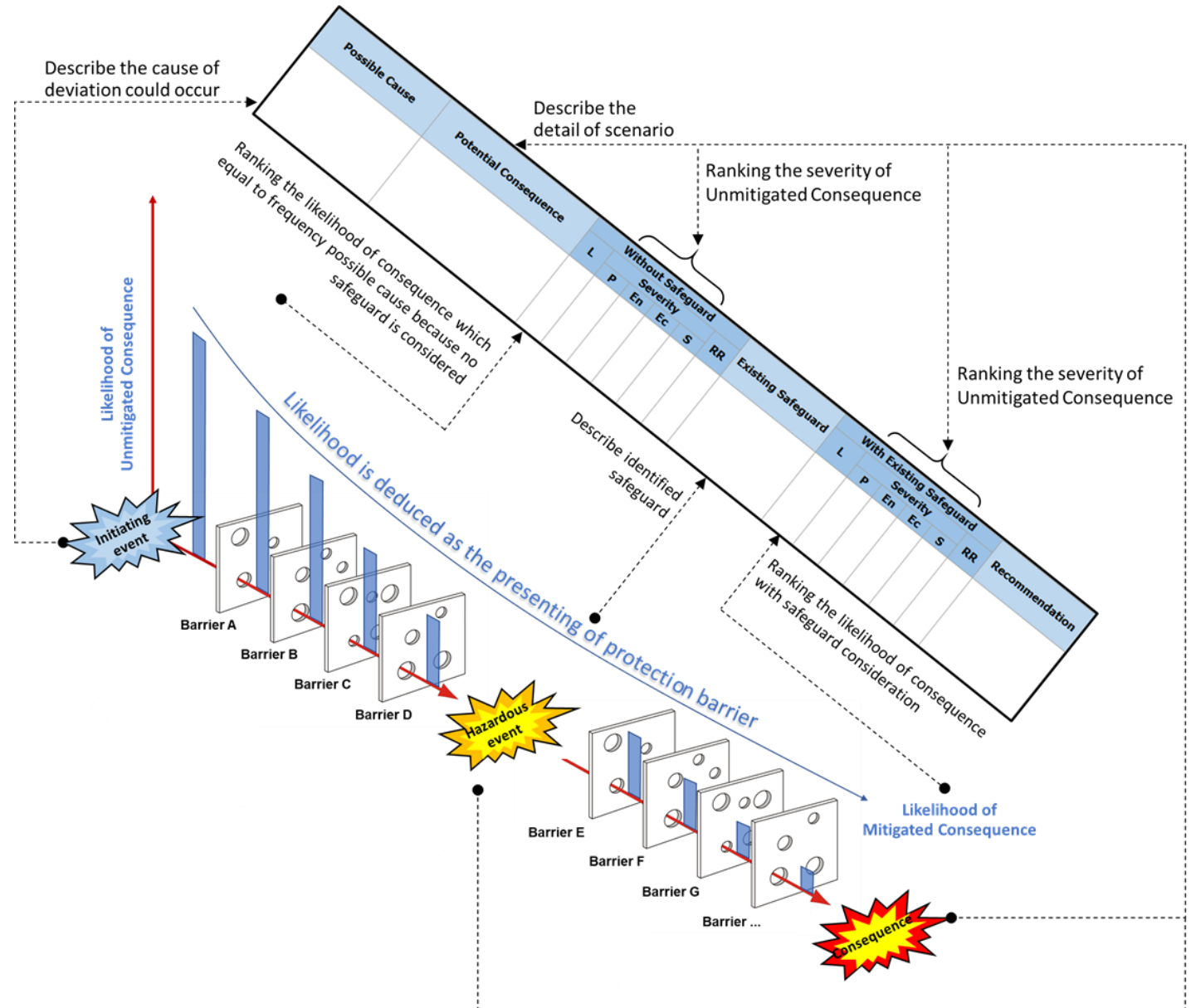
Reduce the occurrence frequency of unmitigated consequence.

Initial Risk (Not consider safeguard)

- Determine the Severity (S) level of the potential consequence
 - Determine the Likelihood (L) of event without consideration of safeguard
- Likelihood (L) is equal to Likelihood of the possible cause.

Mitigated Risk (Consider safeguard)

- Severity of potential consequence are not change*
- Determine the mitigated consequence frequency (Likelihood with safeguard)
- Evaluate the initial risk $S \times L$ (automatic assess)



Indicative guidance for likelihood estimation

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❑ Likelihood estimation of possible cause

- Using the guidance table, if no history of occurrence
- Estimate based on the GC PHA RAM definition, if there are history of occurrence

See additional information in SG-(Q-TS)-014

Possible cause (Initiating Event)	Generic Frequency Range	Guidance of likelihood Level (Qualitative)
Basic process control system (BPCS) instrument loop	1 to 1/10 year	Level 4
Human Error		
• The routine task that is performed once per week or more often (Task $\geq$ once per week)	1 year	Level 5
• The task is performed between once per year and once per week (Once per year $\leq$ Task $<$ once per week)	1 to 1/10 year	Level 4
• The task is performed less often than once a year. (Task $<$ Once per year)	1/10 to 1/100 year	Level 3
Pressure Regulator (Self-Regulator)	1/10 to 1/50 year	Level 4
Pump/ Fan/ Compressor Failure	1 to 1/10 year	Level 4
Loss of Supply from all causes e.g. Upstream supply pump failure, accidental block in from external source (off plot)	1 to 1/10 year	Level 4
Cooling Water Failure	No typical PFD	Refer to the cause of the cooling failure
Site-Wide Loss of Power (Refer to PHA RAM Likelihood definition)		
• If no history of Site-Wide Loss of Power in location.	No typical PFD	Level 3
• If there is history of Site-Wide Loss of Power in location		Level 4
• If there is history of Site-Wide Loss of Power in location more than once per year in location		Level 5

Indicative guidance for reduction likelihood level of IPLs

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❑ IPL Credit and Guidance for likelihood reduction

- Evaluate the safeguard by using IPLs concept
- Using the guidance table to reduce the likelihood

[See additional information in SG-\(Q-TS\)-014](#)

Possible cause (Initiating even)	Generic PFD Range	Guidance likelihood reduction level (Qualitative)
Basic process control system (BPCS) instrument loop	0.1	4 Level
Safety Instrumented Function (SIF) • Achieves SIL1 PFD_{avg} of 0.1 to 0.01 • Achieves SIL2 PFD_{avg} of 0.01 to 0.001 • Achieves SIL3 PFD_{avg} of 0.001 to 0.0001		1 Level 2 Level 3 Level
Alarms with Operator Response (Sufficient time available to respond to the indication of an abnormal condition and complete the required action, and this time is less than the time that it takes for event to become unavoidable)	0.1	1 Level
Routine Operator Surveillance (Sufficient time to indicate the abnormal condition & response and Documented procedures/WI is provide to perform & complete task)	0.1	1 Level
Adjustable movement-limiting device	0.1	1 Level
Pressure Relief Valve (PRV) • PRV must be sized to prevent the scenario consequence • 3 PRV's, where 2 of the 3 PRV's are required to prevent the overpressure and all 3 PRV's are actively always connected to process • 3 PRV's, where 2 of the 3 PRV's are required to prevent the overpressure and all 3 PRV's are actively always connected to the process • More than one PRV is available, but more than one is required to relieve the full load. This includes staged release PRV's	0.01 0.001 0.001 0.1	2 Level 3 Level 3 Level 1 Level

Chapter 4-8

Develop Recommendation

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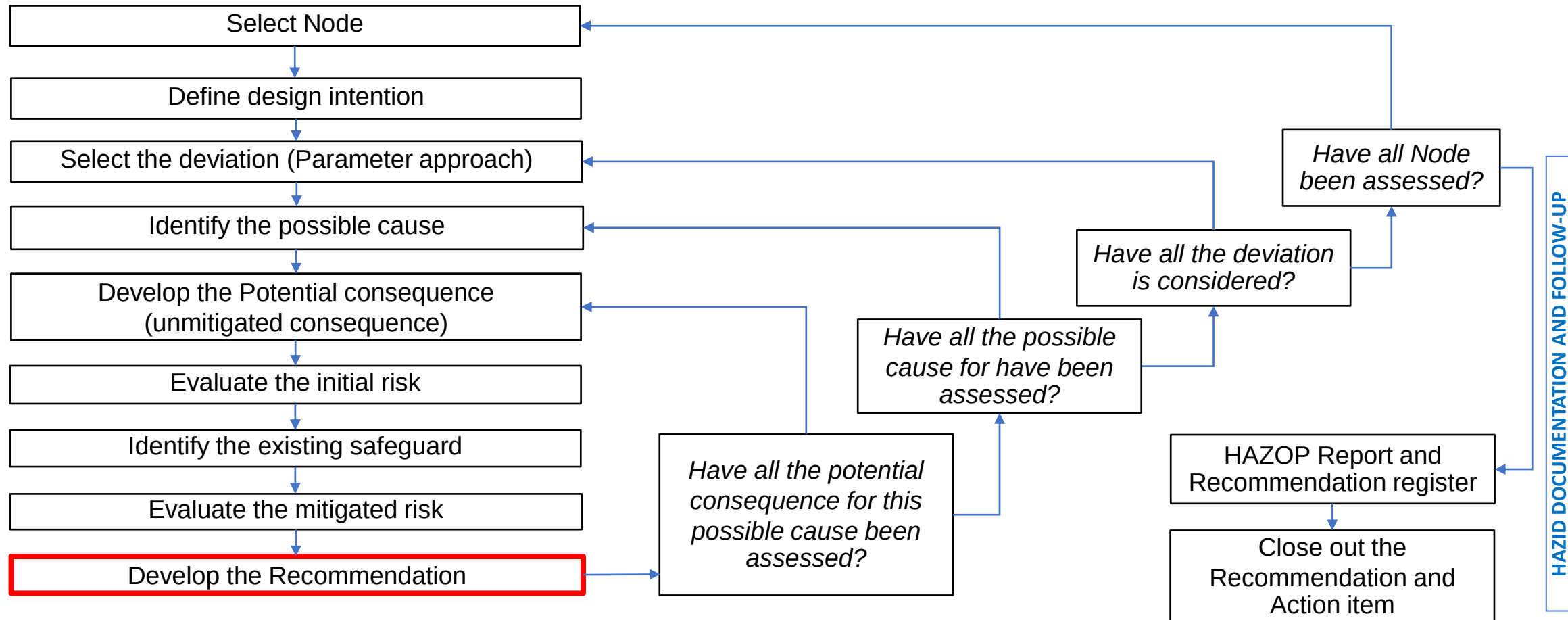
Outline of HAZOP study method

HAZOP Leadership Training Course

HAZOP PREPARATION

1. Define scope & objective study
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4. Agreement for Risk Assessment Matrix (RAM)
5. Process safety information and other support document preparing
6. Node Preparation

HAZOP METHODOLOGY



Develop the recommendation

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- ❑ When mitigated risk is in unacceptable risk level, HAZOP team shall provide the recommendation to reduce the risk to acceptable level.
 - Recommendation may be provided although the mitigated risk already achieves the acceptable level.
 - **Avoid excessive recommendations** : Some HAZOP groups believe that they should issue a recommendation for any scenario that has negative consequences
 - **Avoid Irrelevant recommendations** : Sometimes, people will utilize HAZOP recommendations to obtain approval for an operational or plant design improvement that is not directly related to the safety of personnel or the release of a hazardous chemical.
 - **Trying to solve the recommendation or design the solution during the HAZOP**
The HAZOP can and should result in a list of actions or recommendations, with the designation of someone responsible for carrying them out, but not necessarily the final solution or re-engineering of the plant

Recording of recommendation

- Recommendations should be thoroughly recorded so that a responsible person later can understand the recommendation, its intentions.
- Using the action word (Verb) such as Add, Change, Configure, Provide, Update, Prepare etc.
- Explain the why the recommendation is needed, if possible

Example Recommendation recording

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DON'T

- Add new Level indicator
- Update log sheet
- Add level alarm and drain valve



NO

DO

- Add level transmitter at V-1301 with low level alarm and operator action to prevent cavitation of P310A/B
- Add FI-3128 , FI-3129 , PI-3131 in field operator lock sheet with appropriate value of parameter
- Add high level alarm at E-1235 and drain valve to prevent the hexane contaminate to the fuel gas including prepare the work instruction for draining activity



YES

Chapter 4-9

Worksheet Reference

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❑ Where causes or consequences or recommendations had been previously documented will be referred, HAZOP Leader and HAZOP scribe shall use a specific reference number.

- Do Not record by using “See above”, “See below”, “Same as above”, “Same as earlier node”
- Do not cascade the references, see the following example

Deviation	Causes	Consequences	Safeguards	Recommendations
No Flow	1. ...	1.1 ...	1.1.1 ...	1.1.1
			1.1.2 ...	1.1.2 ...
			1.1.3 ...	
		1.2 ...	1.2.1 ...	
			1.2.2 Same As 1.1.3	
	2. ...	2.1 ...	2.1.1 ...	2.1.1
				2.1.2 Same As 1.1.1
	3. ...	3.1 ...	3.1.1 Same As 1.2.2 Same As 1.1.3	

- Do not refer to numbers in other nodes or other projects. It is better to copy content.

Desired Output

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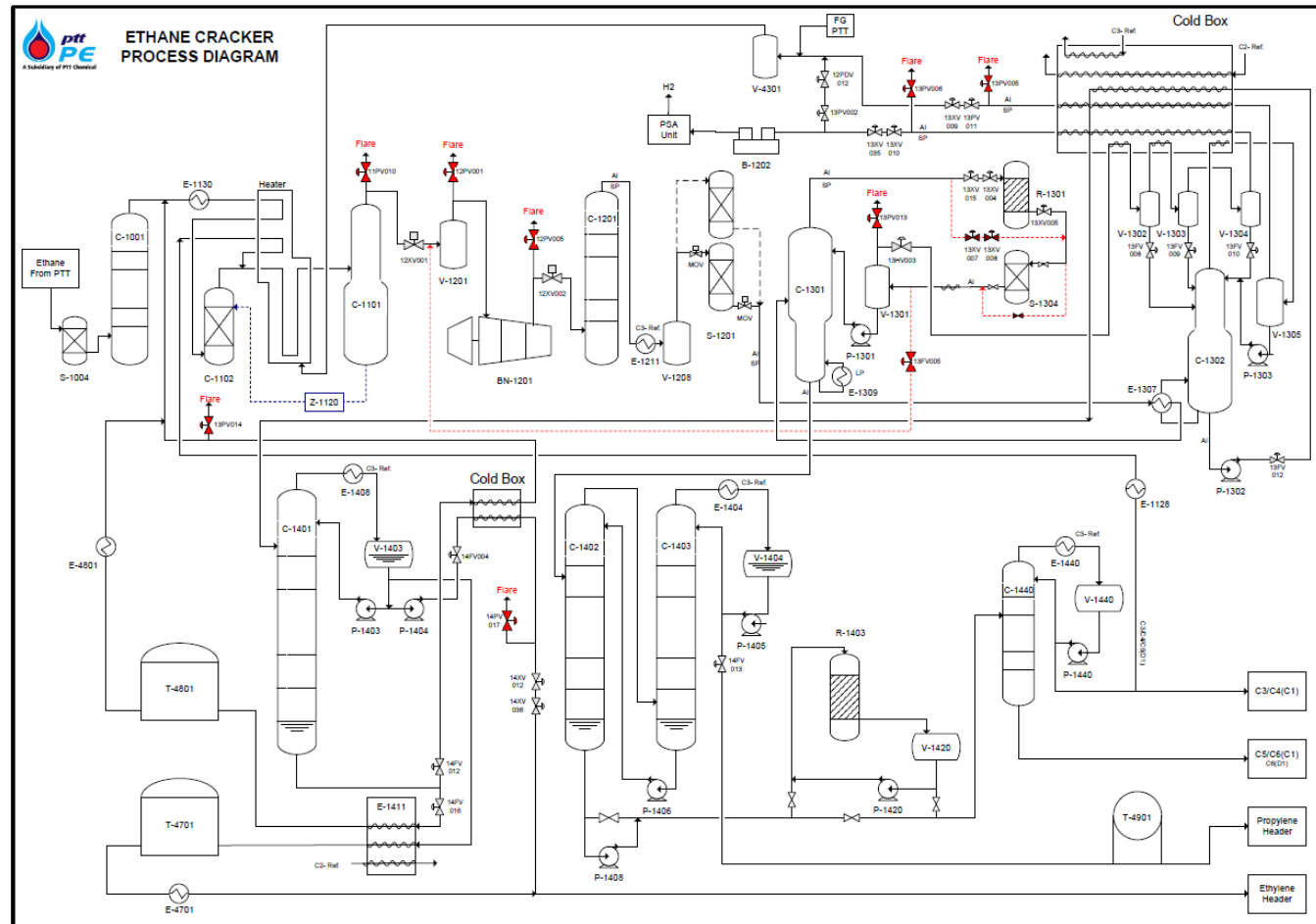
Expected outcome example

Deviation	Cause	Consequence	Existing Safeguard	Recommendation
1. No/Low Flow	1.1. Loss of fuel gas supply (global failure due to closure of control valve 5PV-62016)	1.1.1. Losing supply of fuel gas will result in low pressure in the flare header and potential vacuum condition leading to air ingress. Potential formation of explosive mixture. Potential loss of containment.	1.1.1.1. Low pressure alarm on 5PI- 50014 on HP Flare KO Drum (5V-5001)	10. Provide a parallel letdown control valve redundant to 5PV-62016 on fuel gas supply header.
			1.1.1.2. Back-up N2 supply	
		1.1.2. Flame-off of flare leading to release of hydrocarbon to environment. Potential safety and environmental concern.	1.1.2.1. Low pressure alarm on 5PI- 50014 on HP Flare KO Drum (5V-5001)	10. Provide a parallel letdown control valve redundant to 5PV-62016 on fuel gas supply header.

Activity-8: HAZOP Study (Brake group)

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HAZOP study workshop (All step)



- ☐ Group workshop 4 Group
- ☐ 30 min
- ☐ 10 min for sharing (Each group)

HAZOP Leadership training Course for GC

End of Chapter 4 : HAZOP Methodology
